

Getting Started with phenoscapR

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1 Overview

phenoscapR provides tools for reading, processing, analysing, and visualising multiplexed spatial biology data from Akoya Biosciences platforms (PhenoCycler, CODEX, PhenoImager).

A typical workflow involves:

1. **Reading** cell segmentation data with `read_akoya()`
2. **Quality control** with `qc_filter()`

3. **Normalisation** with `normalise_markers()`
4. **Phenotyping** with `phenotype_cells()`
5. **Spatial analysis** with `nearest_neighbours()`, `cell_density()`, `interaction_matrix()`, and `spatial_clusters()`
6. **Visualisation** with `plot_cell_map()`, `plot_density()`, `plot_heatmap()`, and `plot_interactions()`

2 Creating Example Data

Since real Akoya data files can be large, we will create a small simulated data set for illustration.

```
library(phenoscapR)

set.seed(42)
n_cells <- 500

# Simulate two cell populations in different spatial regions
dt <- data.table::data.table(
  `Cell ID` = seq_len(n_cells),
  `Cell X Position` = c(rnorm(250, 200, 50), rnorm(250, 600, 50)),
  `Cell Y Position` = c(rnorm(250, 300, 50), rnorm(250, 300, 50)),
  `Cell Area (px)` = rlnorm(n_cells, log(100), 0.3),
  CD3 = c(rnorm(250, 800, 150), rnorm(250, 200, 100)),
  CD8 = c(rnorm(250, 200, 100), rnorm(250, 700, 120)),
  DAPI = rnorm(n_cells, 1000, 200)
)

# Write to a temporary CSV
tmp <- tempfile(fileext = ".csv")
write.csv(dt, tmp, row.names = FALSE)
```

3 Reading Data

```
cells <- read_akoya(tmp, sample_id = "example")
head(cells)
```

#>	sample_id	cell_id	x	y	cell_area	CD3	CD8	DAPI
#>	<char>	<int>	<num>	<num>	<num>	<num>	<num>	<num>
#> 1:	example	1	268.5479	351.4570	200.87599	709.7926	225.0578067	1123.4673
#> 2:	example	2	171.7651	345.7387	117.02725	779.6276	172.2075951	999.0918
#> 3:	example	3	218.1564	299.8772	133.80590	651.9091	27.5264266	981.7487
#> 4:	example	4	231.6431	306.8005	111.97350	924.7888	-0.6704944	1079.9919
#> 5:	example	5	220.2134	263.9923	74.17226	680.7411	70.8191670	1117.7803
#> 6:	example	6	194.6938	290.0938	83.59012	851.0697	236.5838228	996.6244

4 Quality Control

```
cells <- qc_filter(cells, min_area = 50, max_area = 500)
nrow(cells)
#> [1] 493
```

5 Normalisation

```
cells <- normalise_markers(cells, method = "zscore")
summary(cells[, c("CD3", "CD8", "DAPI")])
#>      CD3      CD8      DAPI
#> Min.   :-1.6156 Min.   :-1.93580 Min.   :-2.97169
#> 1st Qu.: -0.9104 1st Qu.: -0.92825 1st Qu.: -0.63287
#> Median :-0.2684 Median :-0.05788 Median : 0.05662
#> Mean   : 0.0000 Mean   : 0.00000 Mean   : 0.00000
#> 3rd Qu.: 0.9048 3rd Qu.: 0.90400 3rd Qu.: 0.60278
#> Max.   : 2.5022 Max.   : 2.39231 Max.   : 2.70425
```

6 Phenotyping

```
cells <- phenotype_cells(cells, thresholds = list(CD3 = 0, CD8 = 0))
summarise_phenotypes(cells)
```

7 Spatial Analysis

7.1 Nearest Neighbours

```
cells <- nearest_neighbours(cells, k = 5)
summary(cells$nn_distance)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>  4.794   7.516  10.435  13.540  15.533  89.053
```

7.2 Cell Density

```
cells <- cell_density(cells, radius = 50)
summary(cells$density)
#>      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>   0.00   33.00   55.00   53.76   77.00  103.00
```

7.3 Interaction Matrix

```
interactions <- interaction_matrix(cells, radius = 50)
interactions
```

8 Visualisation

8.1 Cell Map

```
plot_cell_map(cells, point_size = 1)
```

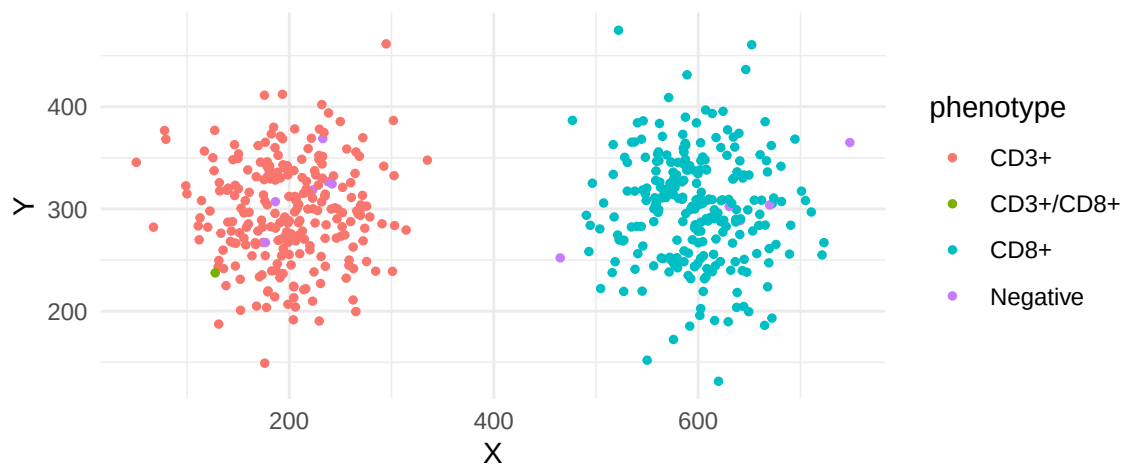


Figure 1: Cell phenotype map

8.2 Density Plot

```
plot_density(cells, point_size = 1)
```

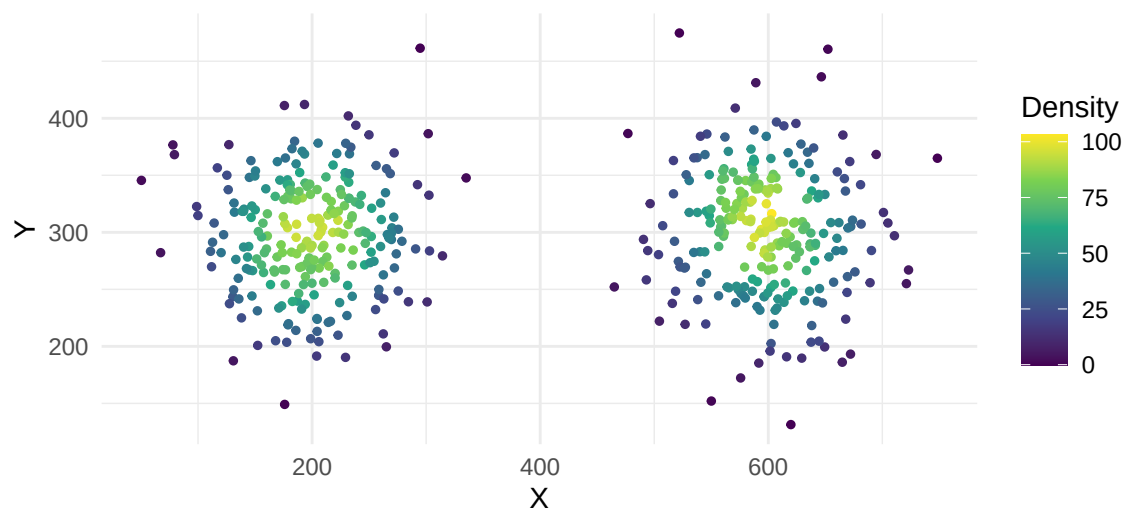


Figure 2: Cell density map

8.3 Marker Heatmap

```
plot_heatmap(cells)
```

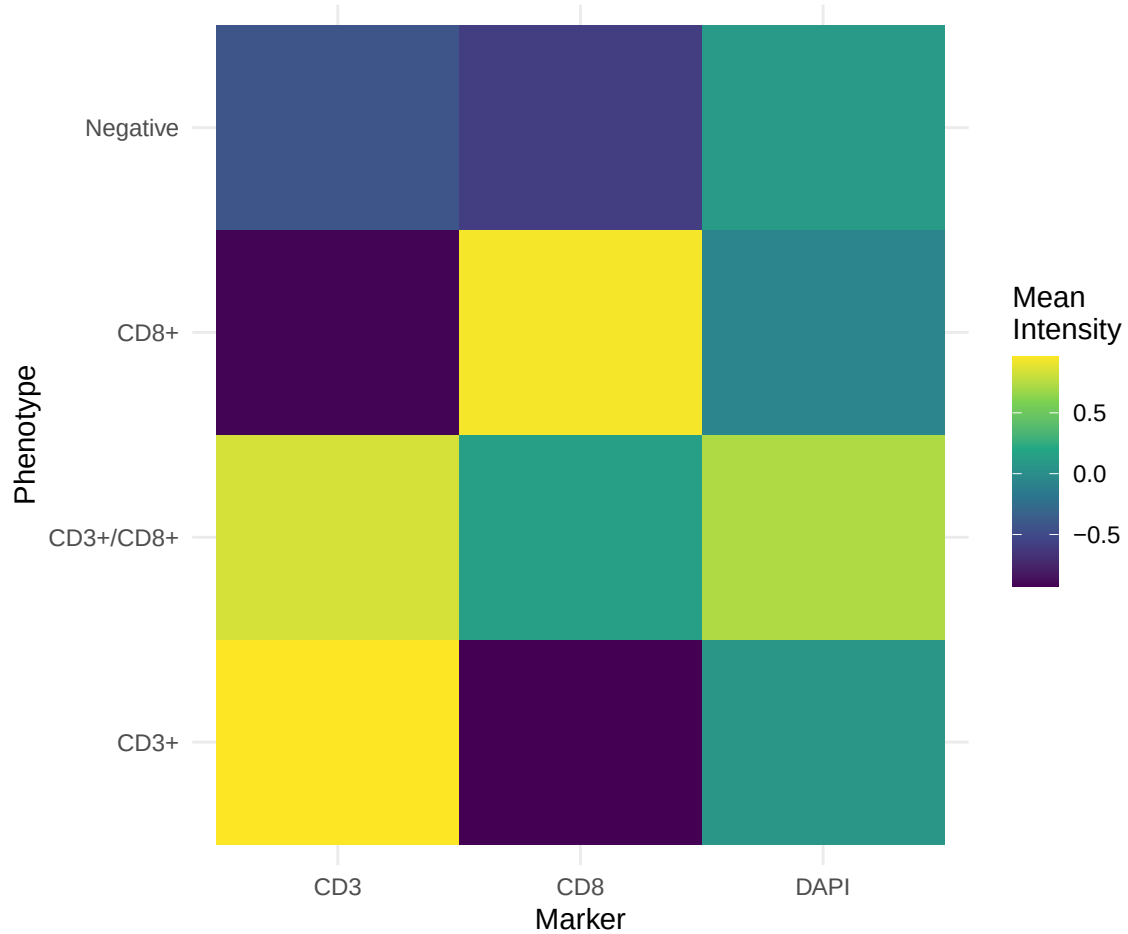


Figure 3: Mean marker expression per phenotype

8.4 Interaction Heatmap

```
plot_interactions(interactions)
```

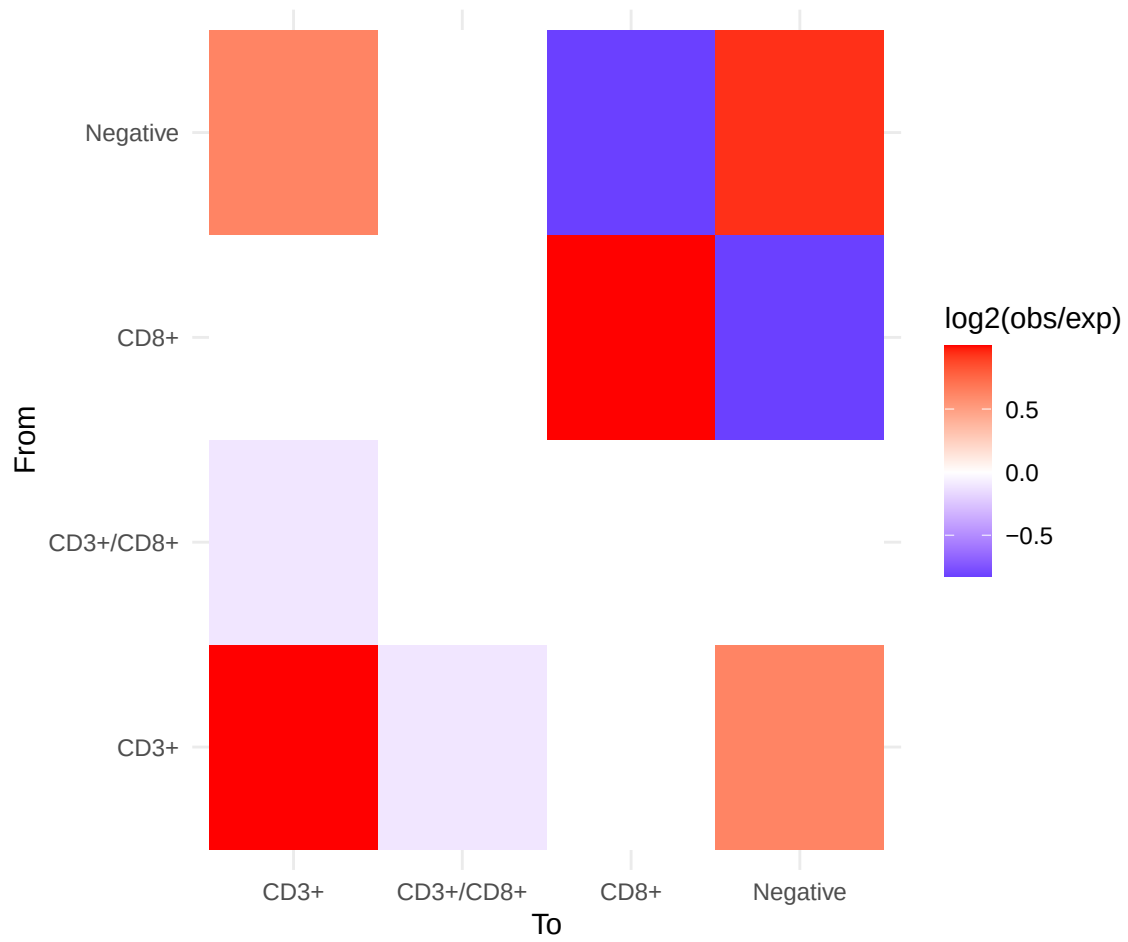


Figure 4: Spatial interaction scores